Question Paper

Exam Date & Time: 18-Nov-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: MAT 2101/MAT_2101 DISCRETE MATHEMATICAL STRUCTURES
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) Draw the Hasse diagram for the Poset $A = \{1, 2, 3, 12, 18, 36\}$ with a partial ordering relation defined as: *a* is related to *b* if and only if *a* divides *b*. Is it a lattice? Justify your answer. Also list all the antichains of length more than one in the Poset A. (4)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

1B) In a Boolean algebra (A, V, Λ, ¯), show that (3)
(i) every element has an unique complement.
(ii) $a \vee (\overline{a} \wedge b) = a \vee b$ and $a \wedge (\overline{a} \vee b) = a \wedge b$ *for all* $a, b \in A$.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

1C) Let $E(x_1, x_2, x_3) = (\overline{x_1} \wedge x_2 \wedge \overline{x_3}) \vee (x_1 \wedge \overline{x_2}) \vee (x_1 \wedge x_3)$ over $(\{0,1\}, \vee, \wedge, \overline{\ })$ be a Boolean expression. Write $E(x_1, x_2, x_3)$ in both CNF and DNF. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

2A) How many ways are there to select a total of 15 items from a collection of three types of items where the selection of each type has the following conditions: (4)
i. Type A selection must be between 1 and 4 items.
ii. Type B selection must be between 3 and 7 items.
iii. Type C selection must be between 5 and 10 items.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

2B) Show that the number of partitions of n in which no part is smaller than 3, is equal to the number of partitions of n+3 in which the three largest parts are consecutive integers. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

2C) Find the 75th permutation of the five marks 0, 1, 2, 3, 4 in Fike's order. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

3A) (4)

(i) Prove that the set D of all odd integers forms an abelian group with respect to the binary operation $*$ defined by $a * b = a + b - 1 \ \forall \ a, b \in D$.

(ii) Let H and K be subgroups of a group G with $HK = KH$. Then prove that HK is a subgroup of G.

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Course Outcome: "CO4" / Cognitive Level: "Application"

3B) (3)

Show that if G is a regular self-complementary graph of order n, then $n = 4t + 1$ for some integer t.

Course Outcome: "CO3" / Cognitive Level: "Application"

3C) (3)

Solve
$2x + y + z = 5$
$3x + 5y + 2z = 15$
$2x + y + 4z = 8$ by LU decomposition method.

Course Outcome: "CO6" / Cognitive Level: "Application"

4A) (4)

Prove that i) Every cyclic group is abelian.
ii) Any group with at most five elements is abelian.

Course Outcome: "CO4" / Cognitive Level: "Application"

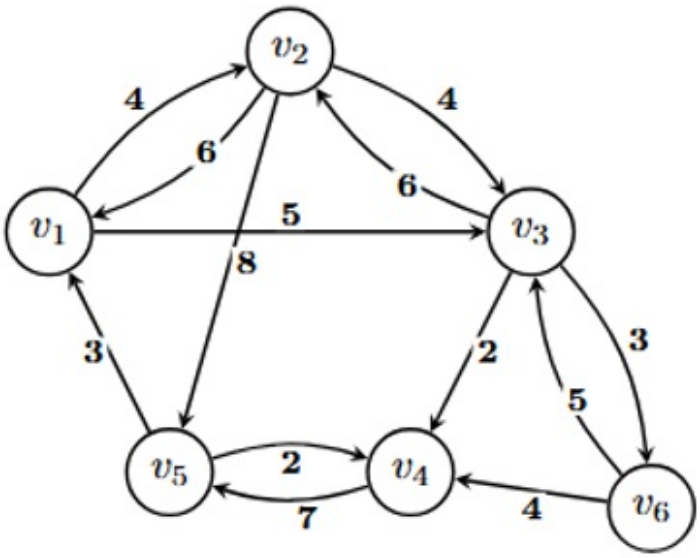
4B) (3)

If G has p vertices and minimum degree $\delta(G) \geq \frac{p-1}{2}$, then prove that G is connected.

Course Outcome: "CO3" / Cognitive Level: "Application"

4C) (3)

Use Dijkstra's algorithm to find the shortest paths from v_1 to all other vertices for the network given below.



Course Outcome: "CO3" / Cognitive Level: "Application"

5A) (4)

Find a singular value decomposition of the matrix $\begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$.

Course Outcome: "CO6" / Cognitive Level: "Application"

5B) (3)

Consider the following premises:
If there was a ball game, then travelling was difficult.
If they arrived on time, then travelling was not difficult.
They arrived on time.
Using propositional inference theory, show that there was no ball game.

Using propositional inference theory, show that there was no ball game.

Course Outcome: "CO5" / Cognitive Level: "Application"

5C) Show that from the premises (3)

$$(\exists y)\left(M(y) \wedge \neg W(y)\right) \text{ and}$$

$$(\exists x)\left(F(x) \wedge S(x)\right) \rightarrow (\forall y)(M(y) \rightarrow W(y))$$

the conclusion $(\forall x)\left(F(x) \rightarrow \neg S(x)\right)$ follows.

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	MAT 2101:Discrete Mathematical Structures

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	9
4	4	CO 4	8
5	5	CO 5	6
6	6	CO 6	7
		Total	**50**

* NA - CO Not Selected

Event Name	Subject
END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	MAT 2101:Discrete Mathematical Structures

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	11	36(72%)
4	Analysis	4	14(28%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)